

SUPPORTLYTICS

Optimizing IT Support Team Performance Using Analytics

Project Report

Student Name: Uma Parvathi Devi Nakireddy

Batch: 1

Guide / Mentor: shanmukha priya

Institution: Pragati Engineering College

Department: Electronics and communication engineering

Degree : B Tech

Academic Year: 4th year

Abstract

Supportlytics is an analytics project developed to convert raw IT support-ticket data into structured, actionable insight for improving IT support team performance. Large organizations generate substantial volumes of support tickets across categories, priorities, channels, regions and teams; without a structured analytical approach it becomes difficult to identify recurring issues, workload imbalances and performance gaps.

The project follows a staged development process. The support-ticket dataset was first reviewed and prepared, covering categorical, numerical and date/time fields such as Priority_score, Customer_Satisfaction, Resolution_Duration and the Created_Date / Resolution_Date pair. Exploratory data analysis was then carried out to understand ticket volume, categories, types, priorities, statuses, channels and regional distribution.

Building on this foundation, a series of visualizations were developed in Python to examine ticket-level patterns, followed by clustering and similarity analysis to group related tickets and evaluate how closely tickets within each group resemble one another. Geographic analysis examined ticket volume and performance across regions and countries, and performance analysis examined resolution duration and satisfaction across departments, teams and priority levels.

The complete set of findings was subsequently consolidated in Microsoft Power BI, resulting in a six-page interactive dashboard — Overview, Tickets, Clusters, Geography, Performance and an executive Report page. This document presents the complete development of Supportlytics organized by project milestone, from Module 1 (project definition) through Module 7 (Power BI dashboard and report), describing what was done, why it was done, how it was done, and the insight each stage was intended to provide. Screenshot placeholders are provided throughout for insertion of the confirmed project outputs.

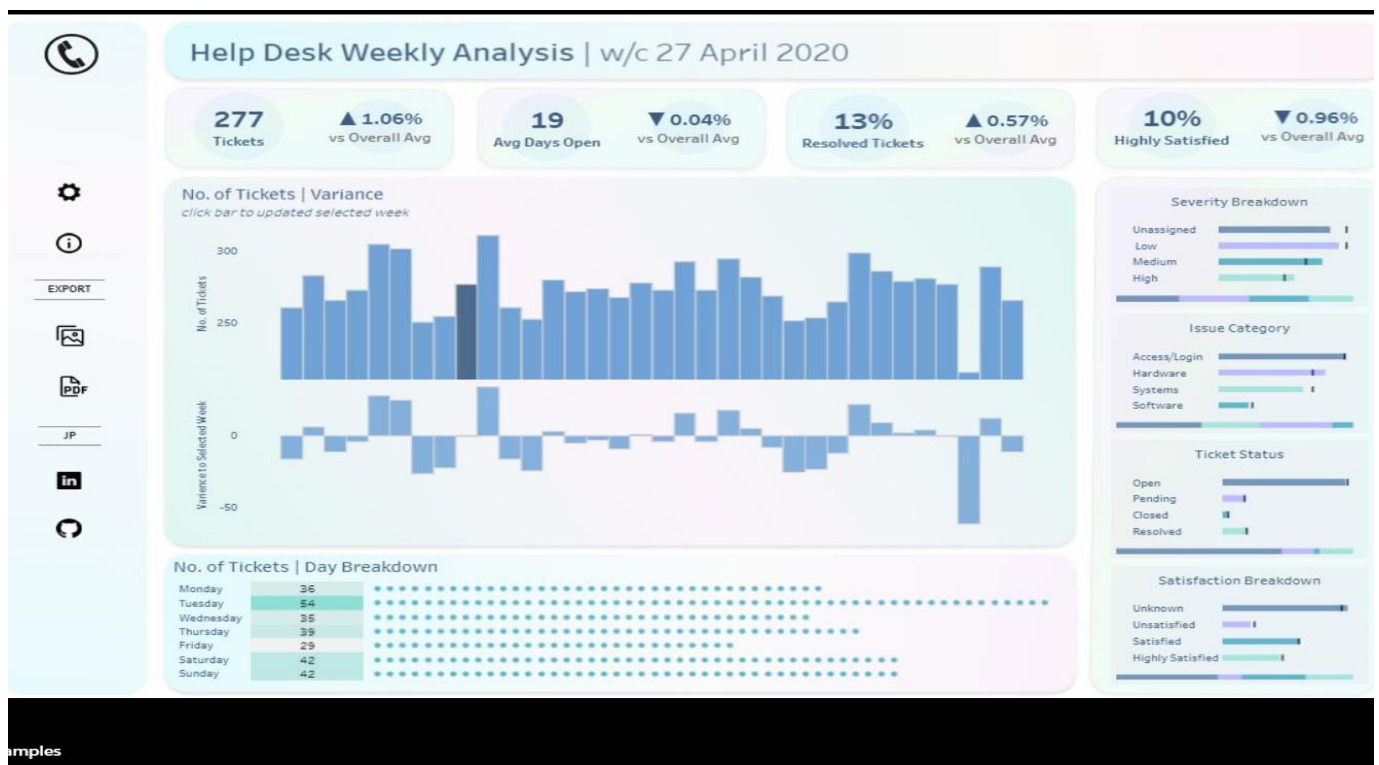


Table of Contents & Project Roadmap

Table of Contents

Abstract 2

Module 1 — Project Introduction & Problem Definition 4

Module 2 — Dataset Understanding & Data Preparation 5

Module 3 — Exploratory Data Analysis 6

Module 4 — Data Visualization & Pattern Analysis 7–8

Module 5 — Clustering & Similarity Analysis 9–10

Module 6 — Geographic & Performance Analysis 11–13

Module 7 — Power BI Dashboard & Report Preparation 14–21

From Module 1 to Module 7 (Project Integration) 22

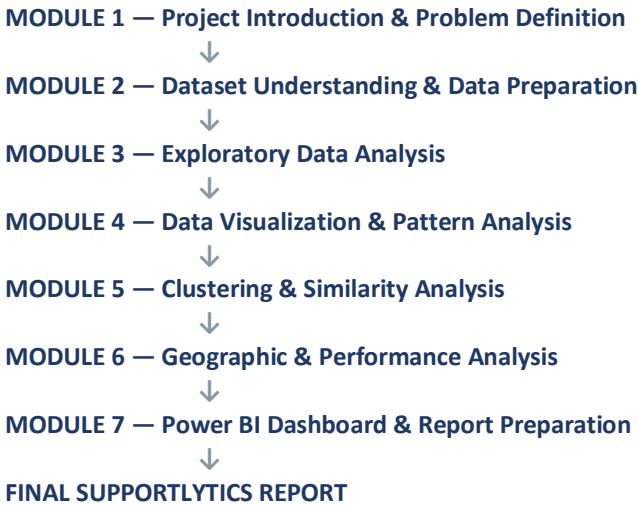
Key Insights & Recommendations 23

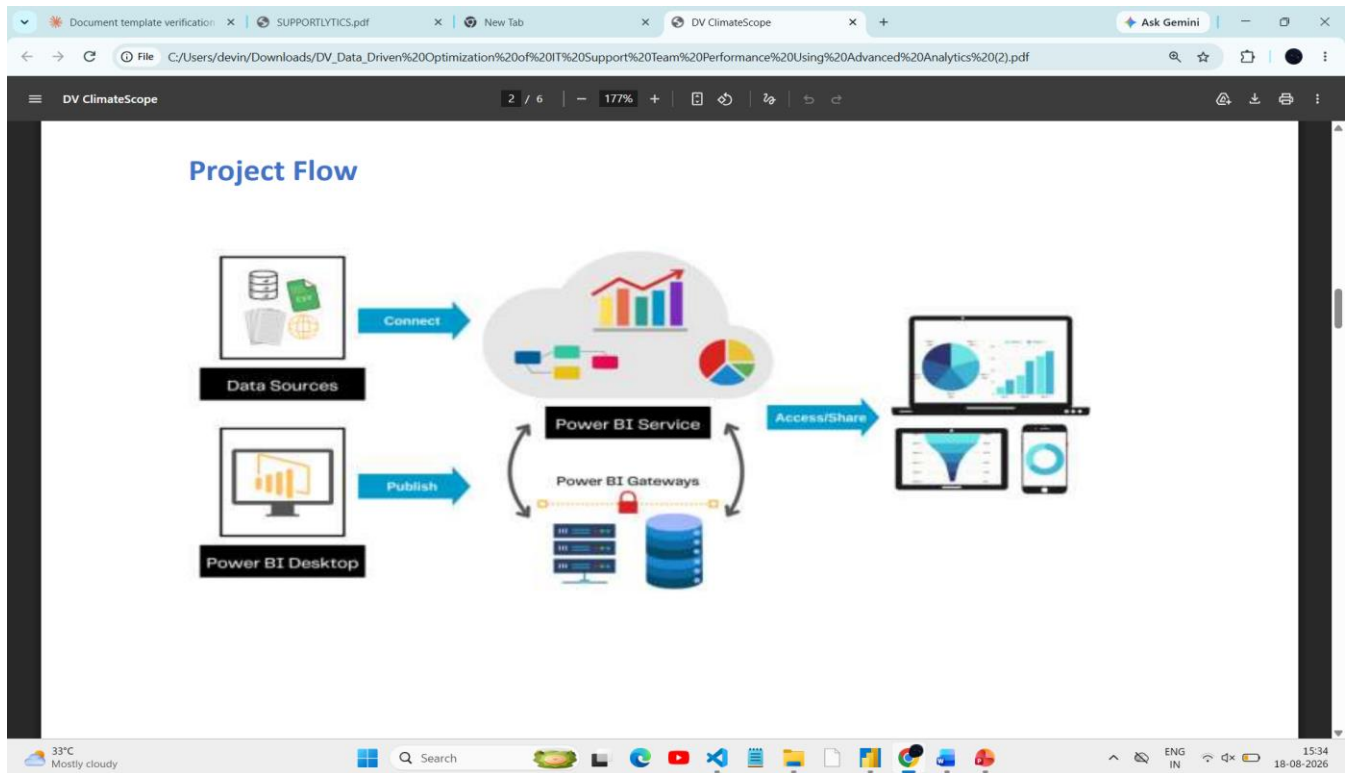
Project Outcome, Limitations & Conclusion 24

Final Dashboard & Appendix 25

Project Roadmap

Each milestone below built directly on the work completed in the previous one, progressively transforming the raw support-ticket dataset into the final interactive Supportlytics dashboard.





Module 1 — Project Introduction & Problem Definition

What Was Done

The initial milestone established the background, problem statement and objectives for the Supportlytics project. This involved reviewing the general context of IT support-ticket analytics and defining the specific problem the project would address.

Why It Was Done

IT support teams handle large volumes of tickets spanning many categories, priorities and channels. Organizations need to analyze this data because, without a structured approach, patterns such as recurring issues, workload imbalances, priority distribution, resolution performance and customer satisfaction differences across regions and teams are difficult to identify manually.

How It Was Done

Work completed during this milestone included reviewing the nature of IT support-ticket data, identifying the analytical problem to be solved, and defining the scope of the project — namely, that ticket volume, issue categories, priorities, resolution time, customer satisfaction, regions and teams would all be examined to improve support operations.

Output Produced

A defined problem statement and an initial set of project objectives, establishing the direction for the dataset preparation and analysis carried out in the following modules.

Contribution to the Final Project

Module 1 established the analytical foundation of Supportlytics: the reason the project exists, the problem it addresses, and the general scope that all subsequent modules — from data preparation through the final Power BI dashboard — were built to satisfy.

Module 2 — Dataset Understanding & Data Preparation

Work completed during this milestone focused on understanding the structure of the support-ticket dataset and preparing it for analysis.

What Was Done / How It Was Done

- Loaded the Excel dataset and reviewed all available columns.
- Checked data types for each field against expected categorical, numerical or date/time formats.
- Prepared categorical fields (Status, Priority, Ticket_Type, Category, Country, Region, Team, Department, Channel).
- Prepared numerical fields (Priority_score, Customer_Satisfaction) and the Resolution_Duration measure.
- Prepared date/time fields, Created_Date and Resolution_Date.
- Prepared the similarity and clustering fields — Cluster_ID, Cluster_Name, Similarity_Score, Similarity_Level.
- Prepared the finalized dataset for Python-based analysis and later import into Power BI.

Dataset Fields

Field	Description	Type / Role
Ticket_ID	Unique identifier for each ticket.	Identifier
Created_Date / Resolution_Date	Date-time ticket was created / resolved.	Date/Time
Status / Priority / Priority_score	Ticket status, priority label and numeric score.	Categorical / Whole No.
Ticket_Type / Category	Type of ticket and issue category.	Categorical
Country / Region	Geographic origin of the ticket.	Categorical
Team / Department / Channel	Handling team, department and support channel.	Categorical
Customer_Satisfaction	Recorded satisfaction score.	Whole Number
Resolution_Duration	Time taken to resolve the ticket.	Decimal Number
Cluster_ID / Cluster_Name	Assigned cluster identifier and name.	Categorical
Similarity_Score / Similarity_Level	Similarity measure and its categorical grouping.	Decimal / Categorical

Contribution to the Final Project

Module 2 provided the data foundation for the entire Supportlytics project. The prepared and validated dataset ensured that the subsequent analysis, visualizations, clustering, geographic analysis, performance analysis, and Power BI dashboard were based on consistently structured and correctly typed data.

🔍 **Established a reliable dataset structure** by organizing the ticket, customer, priority, resolution, geographic, and clustering-related fields for consistent analysis.

🔍 **Enabled accurate analytical calculations** by preparing numerical, categorical, and date/time fields in appropriate formats for later visualizations and KPI development

🔍 **Created the foundation for the Power BI report** by ensuring the dataset was ready for exploratory analysis, clustering analysis, performance evaluation, and interactive dashboard development.

Module 3 — Exploratory Data Analysis

Work completed during this milestone involved an initial exploratory review of the prepared dataset to understand its overall structure before deeper analysis was undertaken.

What Was Examined

- Ticket volume — the overall number of tickets in the dataset.
- Ticket categories and ticket types — how tickets are distributed across each.
- Priority and status — the distribution of tickets across priority levels and status values.
- Support channel — the distribution of tickets across channels.
- Regions — the distribution of tickets geographically.
- Resolution duration and customer satisfaction — the general characteristics of each.

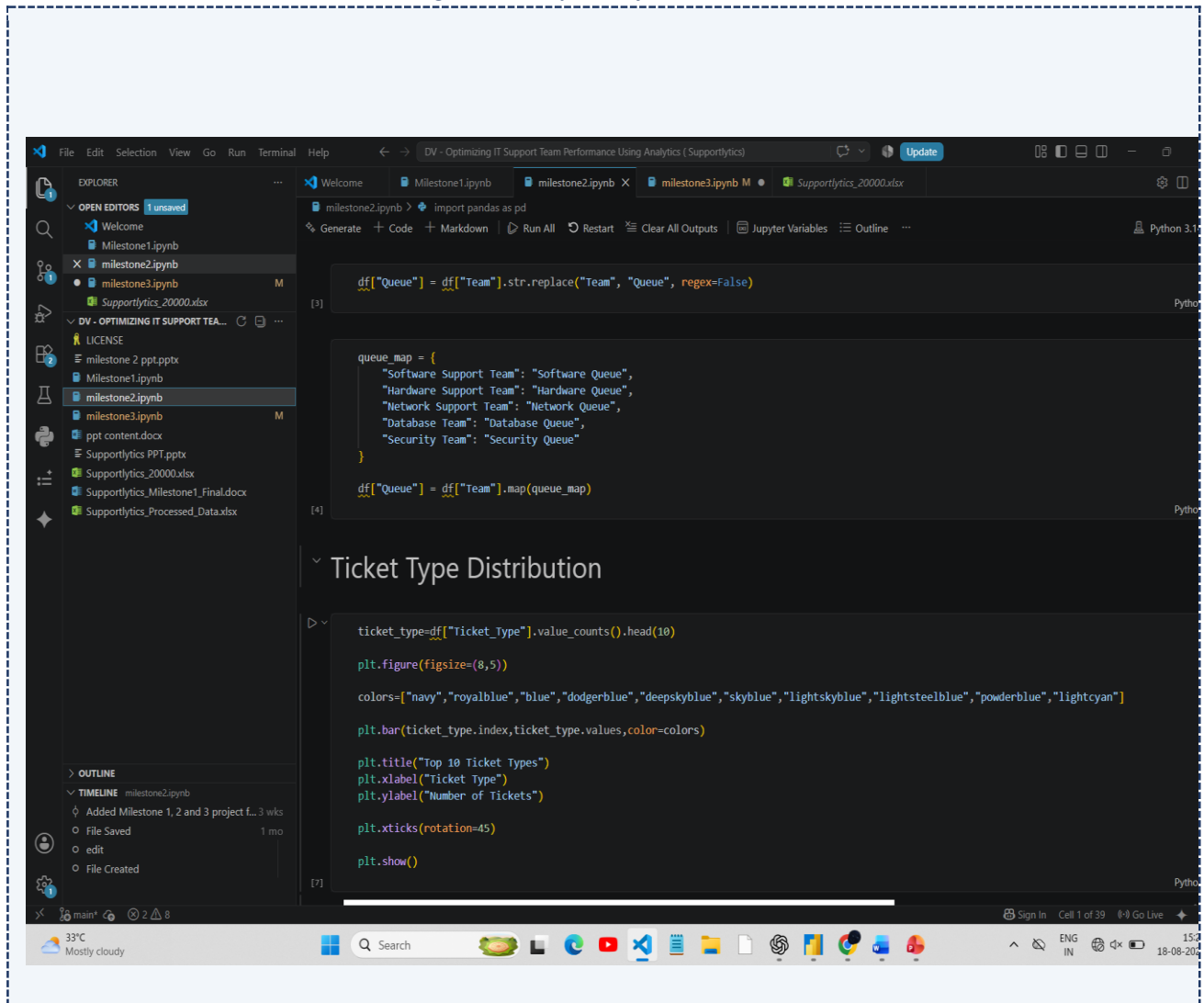
Why It Was Done

Exploratory analysis was performed before creating the final dashboard because it allowed the overall shape of the dataset to be understood first, so that the visualizations and analytical groupings developed later would be meaningful and grounded in the actual characteristics of the data.

Contribution to the Final Project

This milestone identified the main patterns worth visualizing and directly informed the choice of visualizations developed in Module 4 and the more advanced clustering, geographic and performance analysis carried out in Modules 5 and 6.

Figure 3.1 — Exploratory Data Overview



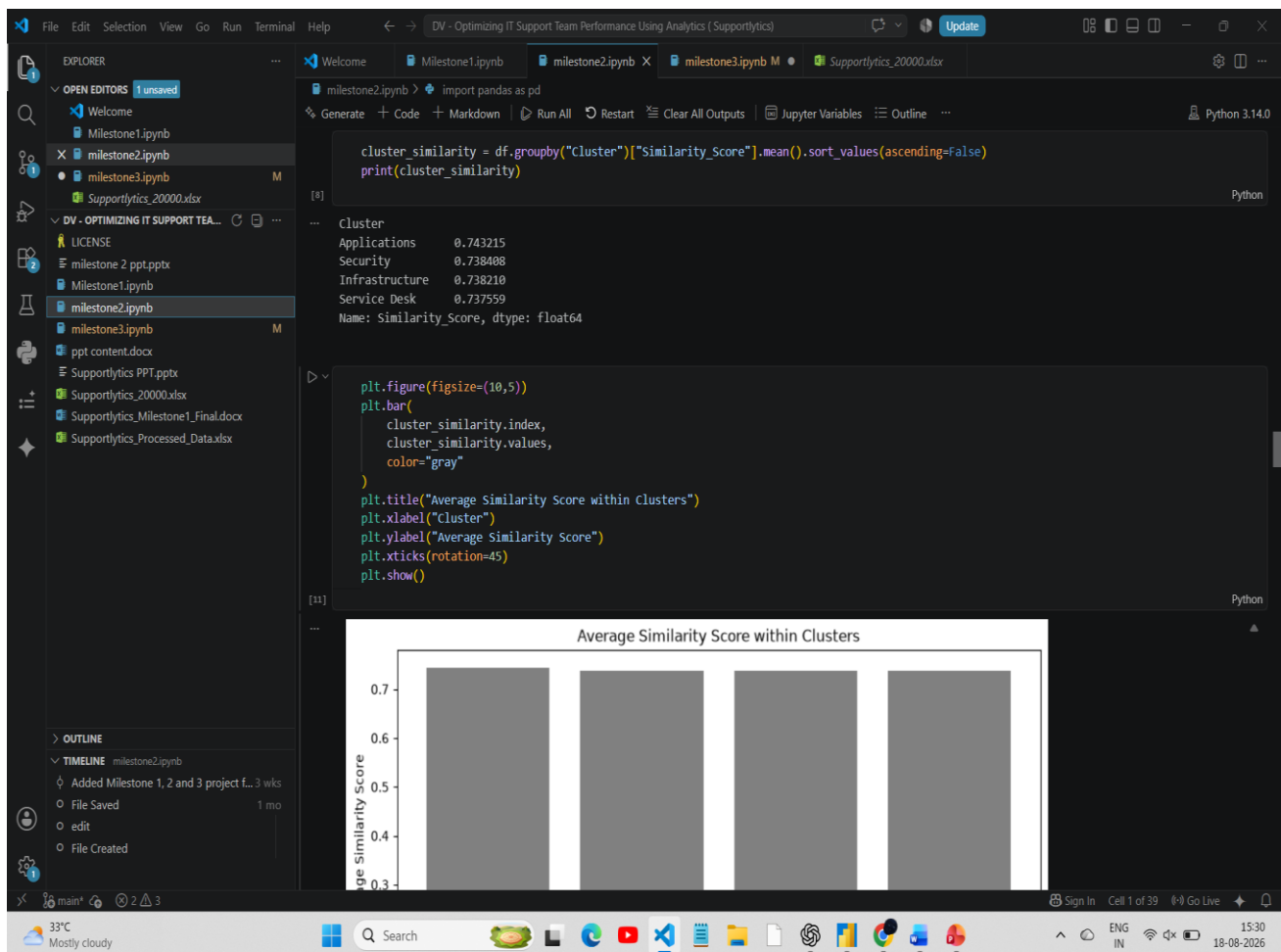
Module 4 — Data Visualization & Pattern Analysis

Work completed during this milestone involved developing Python visualizations of the ticket-level and regional patterns identified during exploratory analysis.

Visualization	What it Represents	Why it is Useful
Ticket Type Distribution	Frequency of each ticket type.	Identifies the most common types of tickets raised.
Issue Category Distribution	Frequency of tickets per issue category.	Highlights categories generating the highest support volume.
Priority Distribution	Proportion of tickets per priority level.	Shows the overall urgency profile of the workload.
Status Distribution	Proportion of tickets per status value.	Shows how much of the workload remains open versus resolved.
Support Channel Analysis	Ticket volume by support channel.	Identifies the channels used most frequently.
Region × Category Analysis	Issue categories compared across regions.	Identifies where specific issue types are concentrated.
Region × Priority Analysis	Ticket concentration by region and priority.	Identifies regions with greater high-priority workload.

Contribution to the Final Project

These visualizations formed the basis of the Tickets-related pages later recreated in Power BI (Module 7), and the region-based comparisons carried directly into the Geography analysis in Module 6.



Module 4 — Visualization Outputs

The following placeholders correspond to the Python visualizations described on the previous page.

Figure 4.1

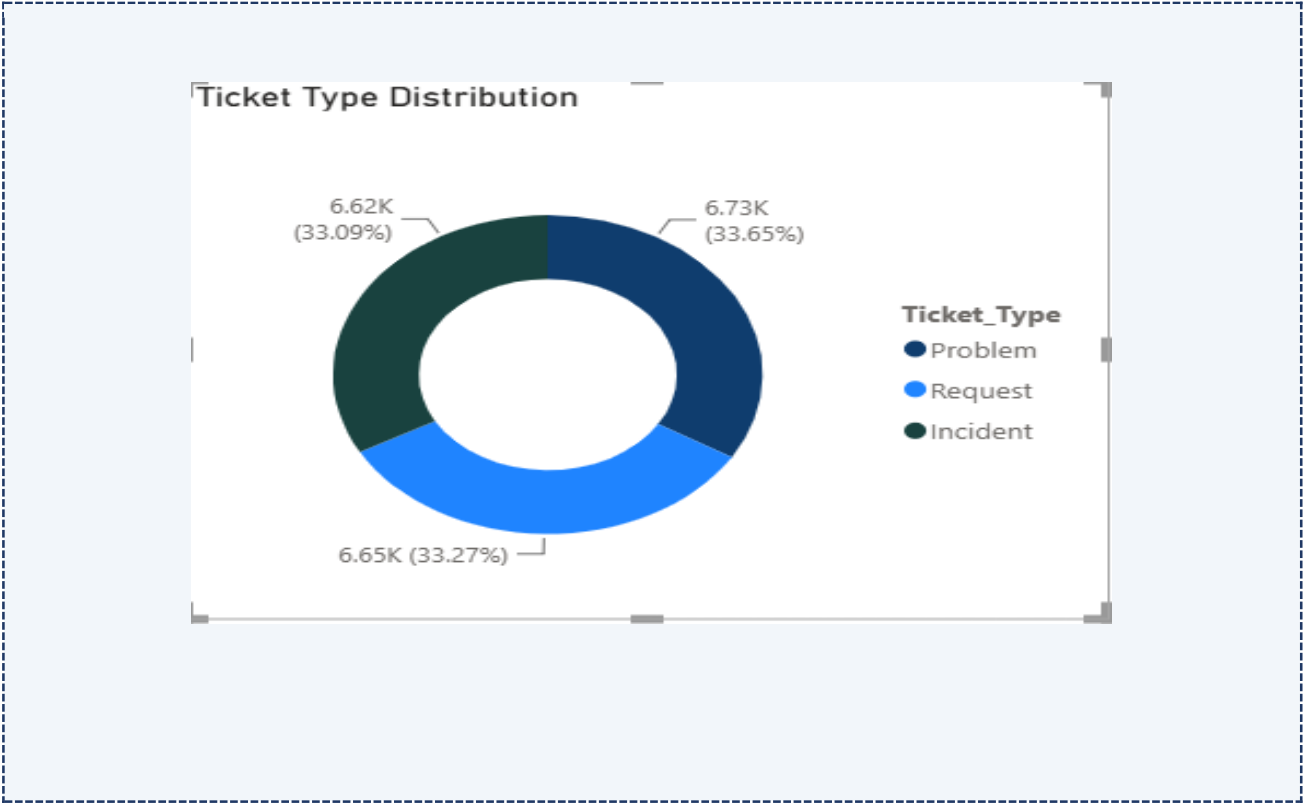


Figure 4.2

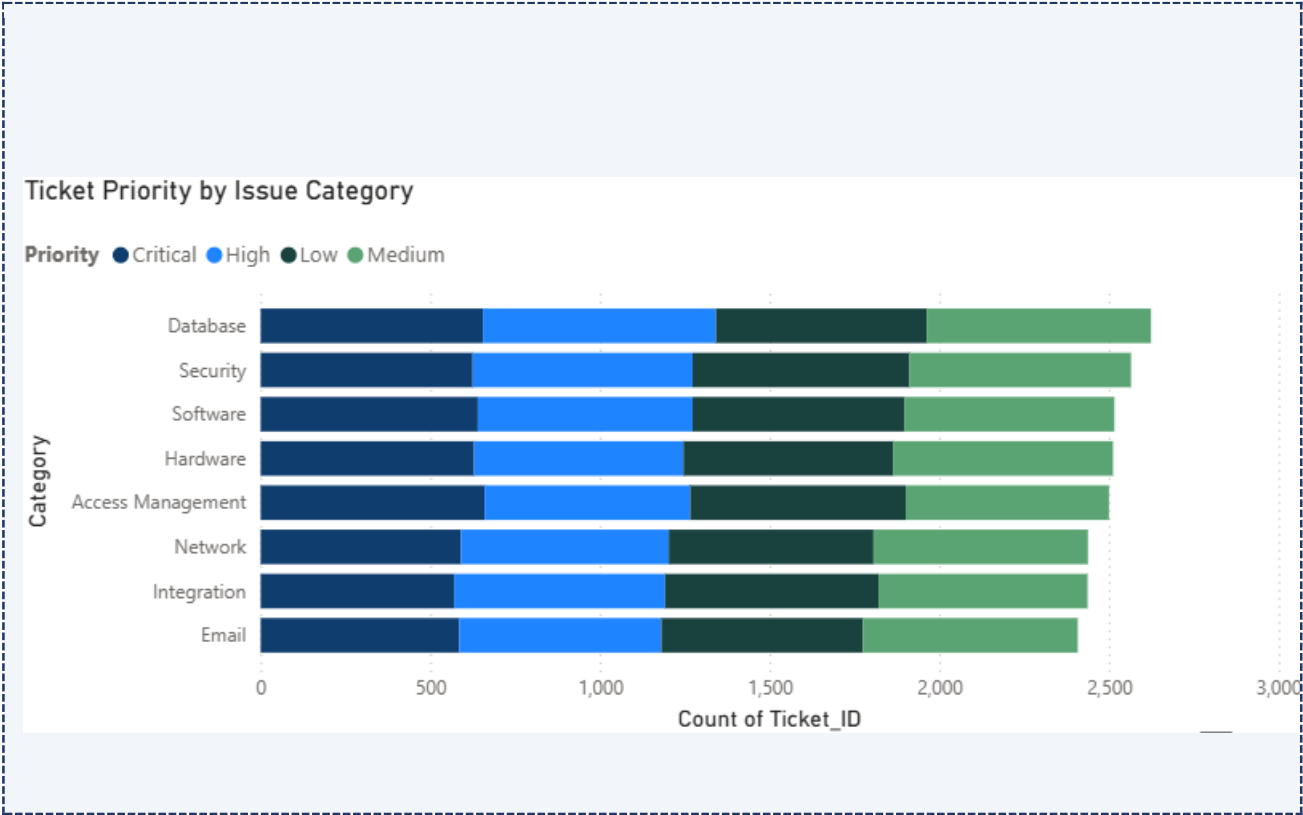
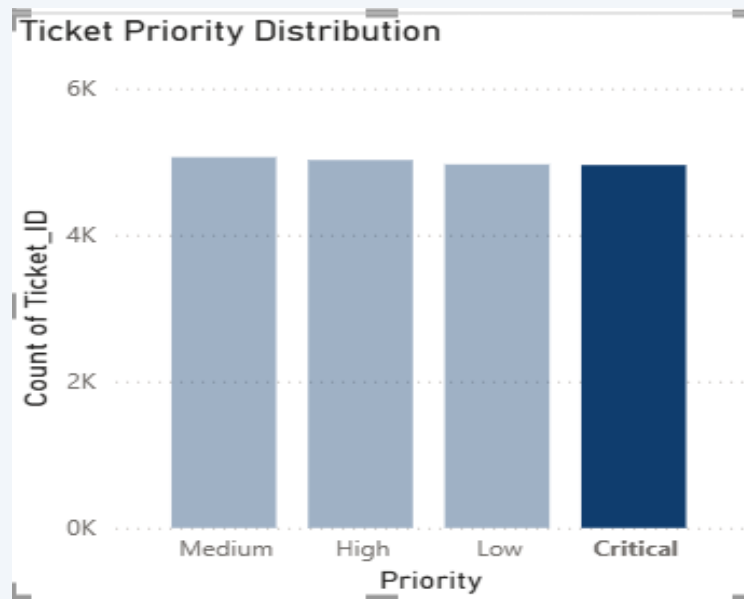
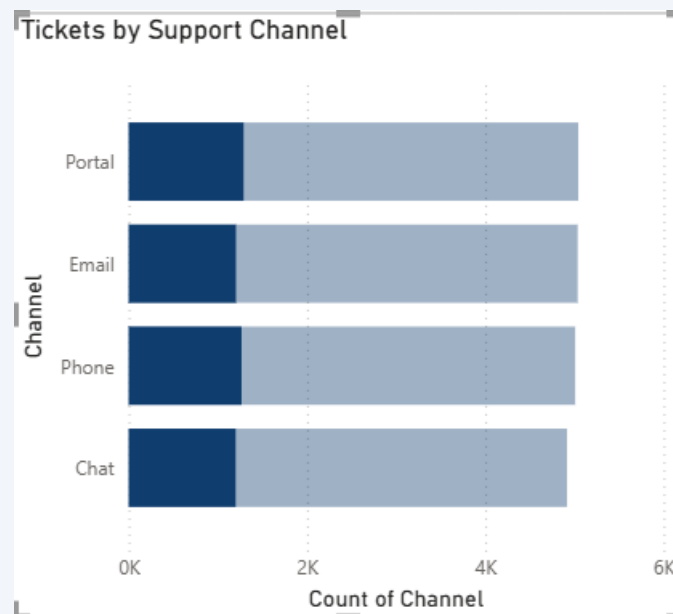


Figure 4.3



*Figure 4.4*

Module 5 — Clustering & Similarity Analysis

Work completed during this milestone applied clustering to group support tickets with similar characteristics, in order to identify recurring patterns not easily visible when tickets are reviewed individually. Similarity analysis was used alongside clustering to evaluate how closely related the tickets within each cluster are.

Why It Was Done

Grouping tickets into clusters allows recurring issue patterns to be identified at a level above individual tickets or categories, and the similarity score provides a way to assess how coherent each identified group is.

Cluster Analyses Produced

- Ticket Distribution Across Clusters — how tickets are distributed among the identified clusters.
- Average Similarity Score by Cluster — the average similarity score within each cluster.
- Issue Category Distribution Across Clusters — how issue categories are represented within each cluster.
- Ticket Volume by Cluster — the total number of tickets in each cluster.
- Overall Cluster Similarity Index — a summary measure of similarity/coherence across all clusters.

Specific similarity values are not stated in this report and should be confirmed from the completed clustering results.

Contribution to the Final Project

The cluster and similarity outputs developed in this module were recreated directly on the Clusters page of the Module 7 Power BI dashboard (Page 17).

Module 5 — Cluster & Similarity Outputs

Figure 5.1

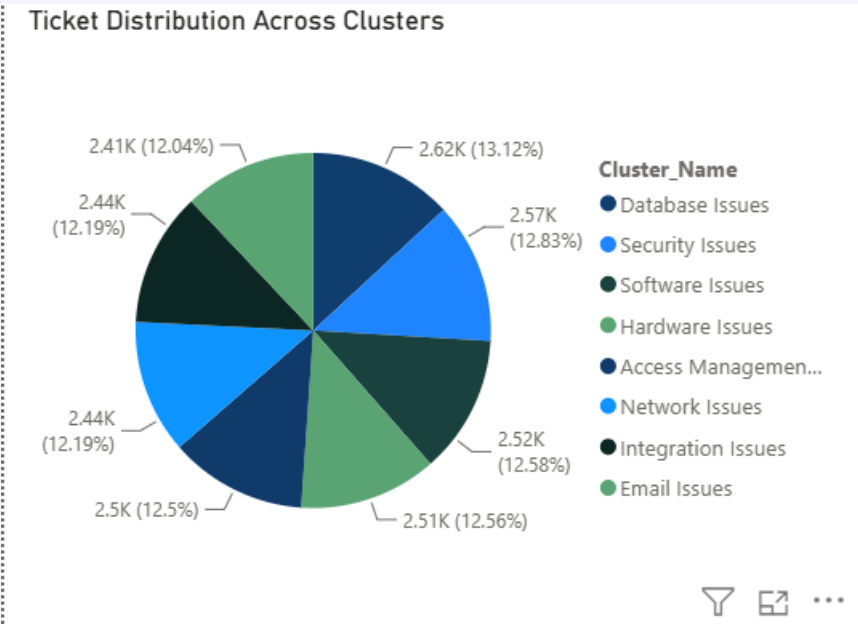
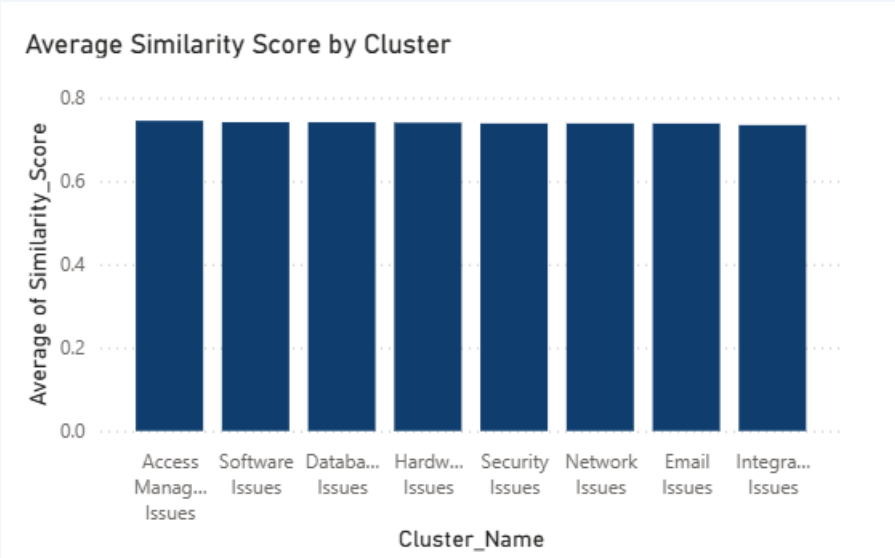


Figure 5.2



CLUSTER SIMILARITY INDEX

0.74

Average of Similarity_Score

Module 6 — Geographic & Performance Analysis

Work completed during this milestone extended the analysis to geographic and performance dimensions ahead of the final Power BI dashboard.

Geographic Analysis

- Regional Ticket Distribution — ticket volume by region.
- Average Resolution Time by Region — resolution duration compared across regions.
- Ticket Distribution by Country — ticket volume at country level.
- Customer Satisfaction by Region — satisfaction compared across regions.
- Regional Performance — a combined comparison of regional support performance.

Performance Analysis

- Average Resolution Time by Department — resolution duration compared across departments.
- Average Customer Satisfaction by Team — satisfaction compared across teams.
- Average Resolution Time by Priority — resolution duration compared across priority levels.
- Ticket Status by Priority — relationship between priority and current status.
- Average Similarity Score by Team — links team-level analysis to clustering results.

Contribution to the Final Project

These analyses were later recreated on the Geography and Performance pages of the Module 7 Power BI dashboard (Pages 18–19).

🔍 **Extended the analysis beyond ticket-level patterns** by introducing geographic and regional perspectives into the project.

🔍 **Enabled regional performance comparison** by analyzing ticket volume, resolution time and customer satisfaction across different regions.

🔍 **Added department and team-level performance analysis** to support a more detailed understanding of operational efficiency.

🔍 **Connected performance metrics with priority and ticket status**, helping provide a broader view of support workload and resolution performance.

🔍 **Prepared the analytical foundation for the final Power BI dashboard**, where the geographic and performance analyses were incorporated into the Geography and Performance pages of Module 7.

Module 6 — Geographic Analysis Outputs

Figure 6.1

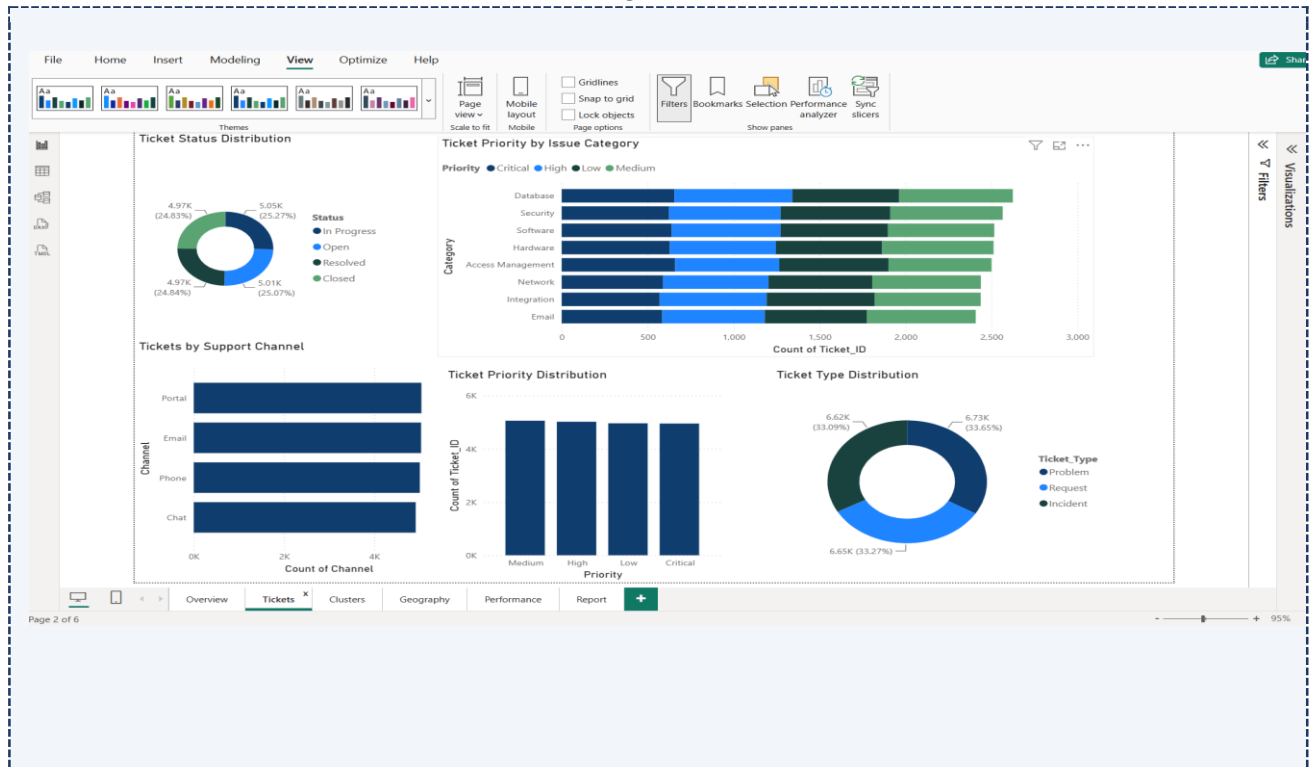


Figure 6.2

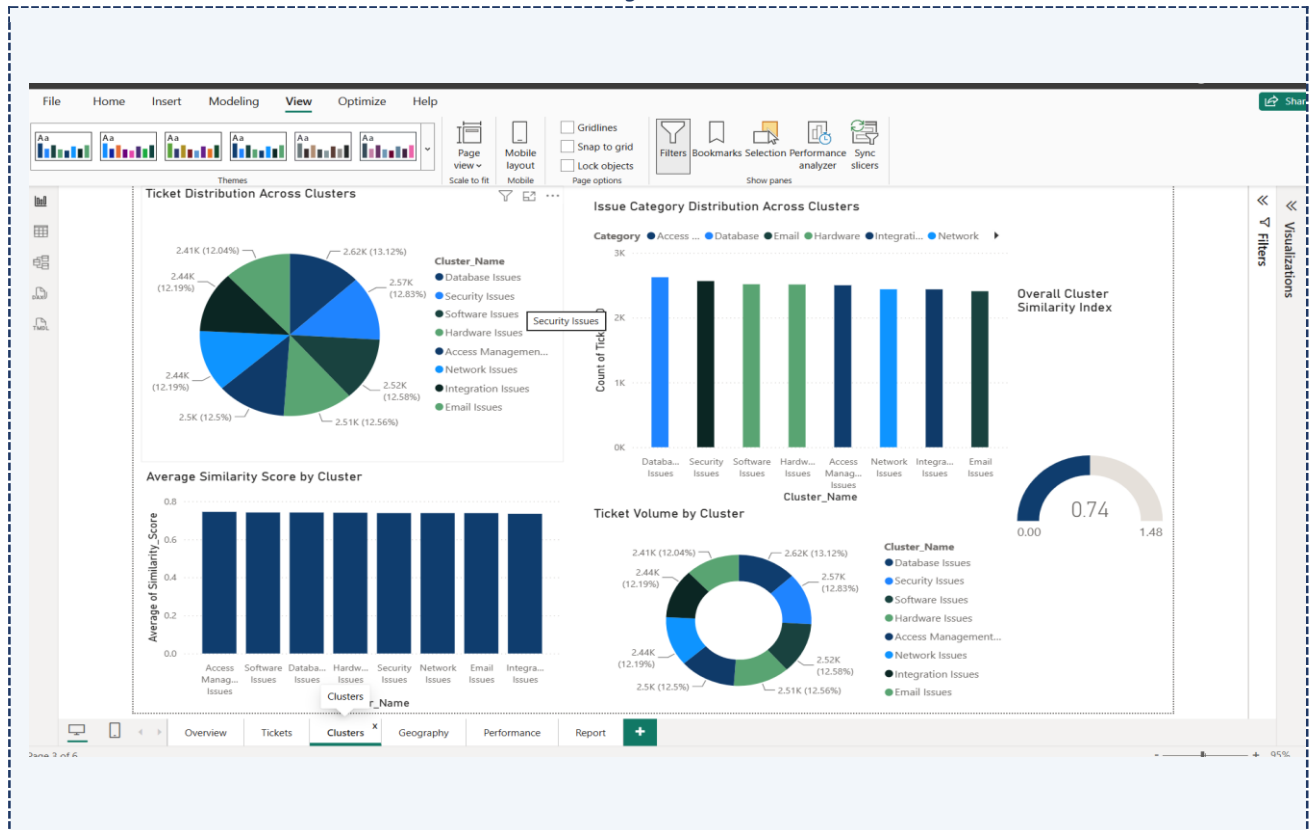


Figure 6.3

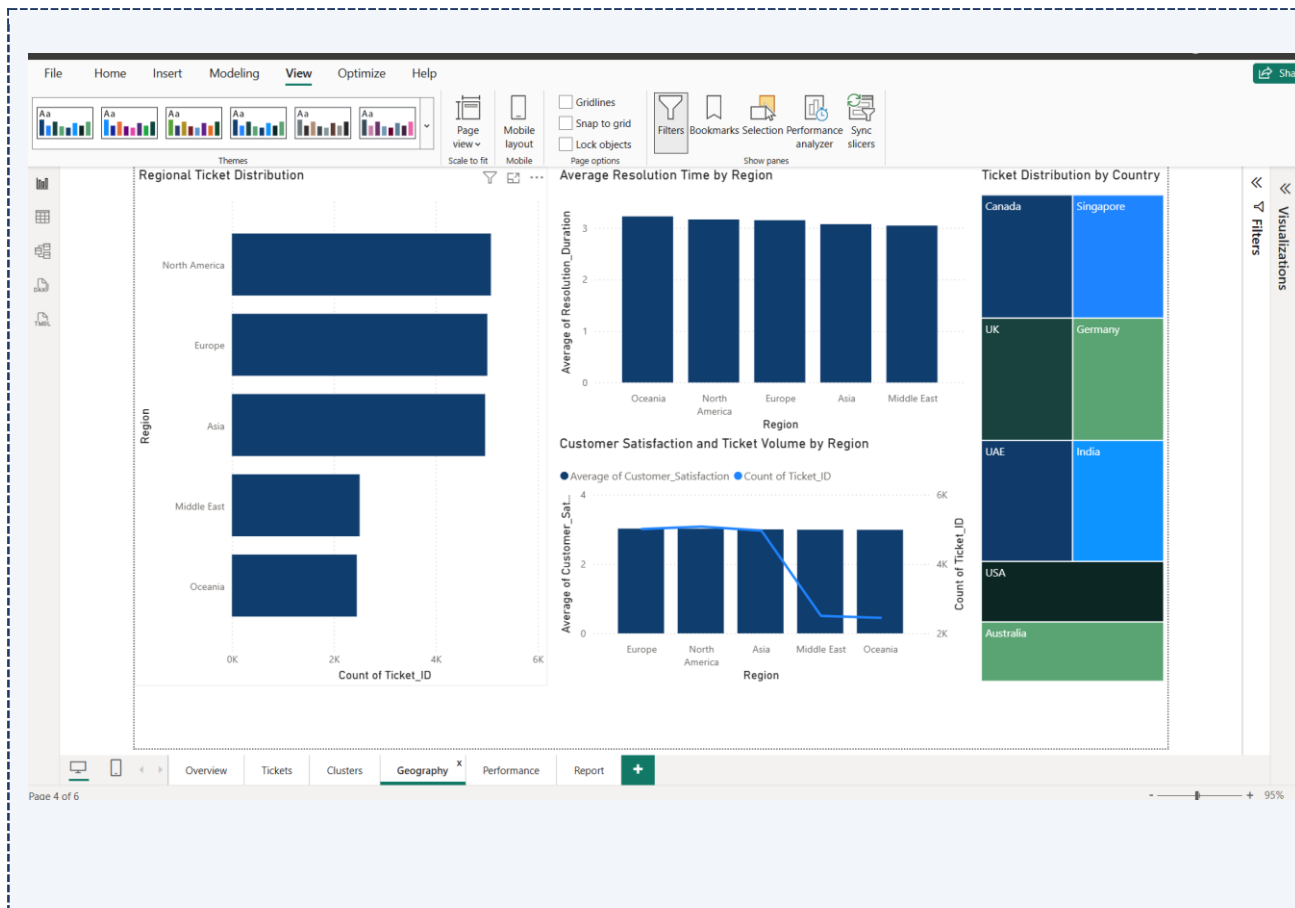


Figure 6.4

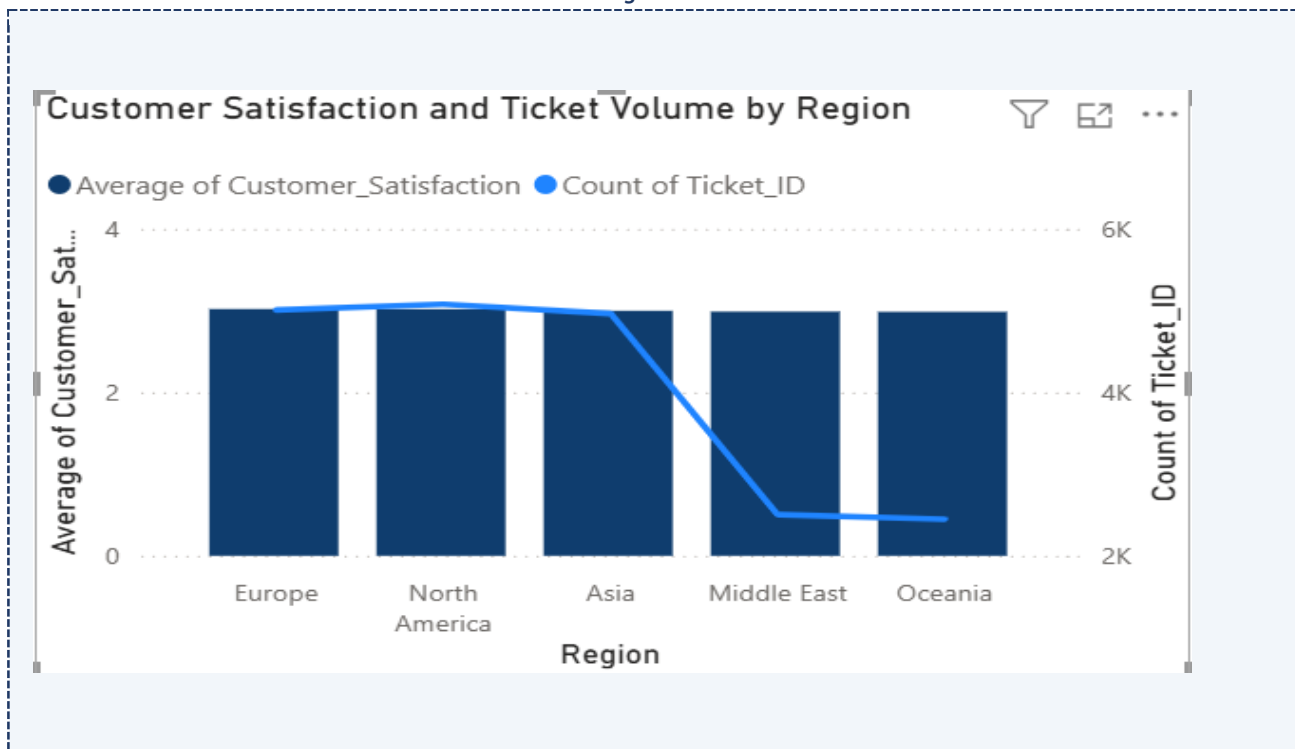
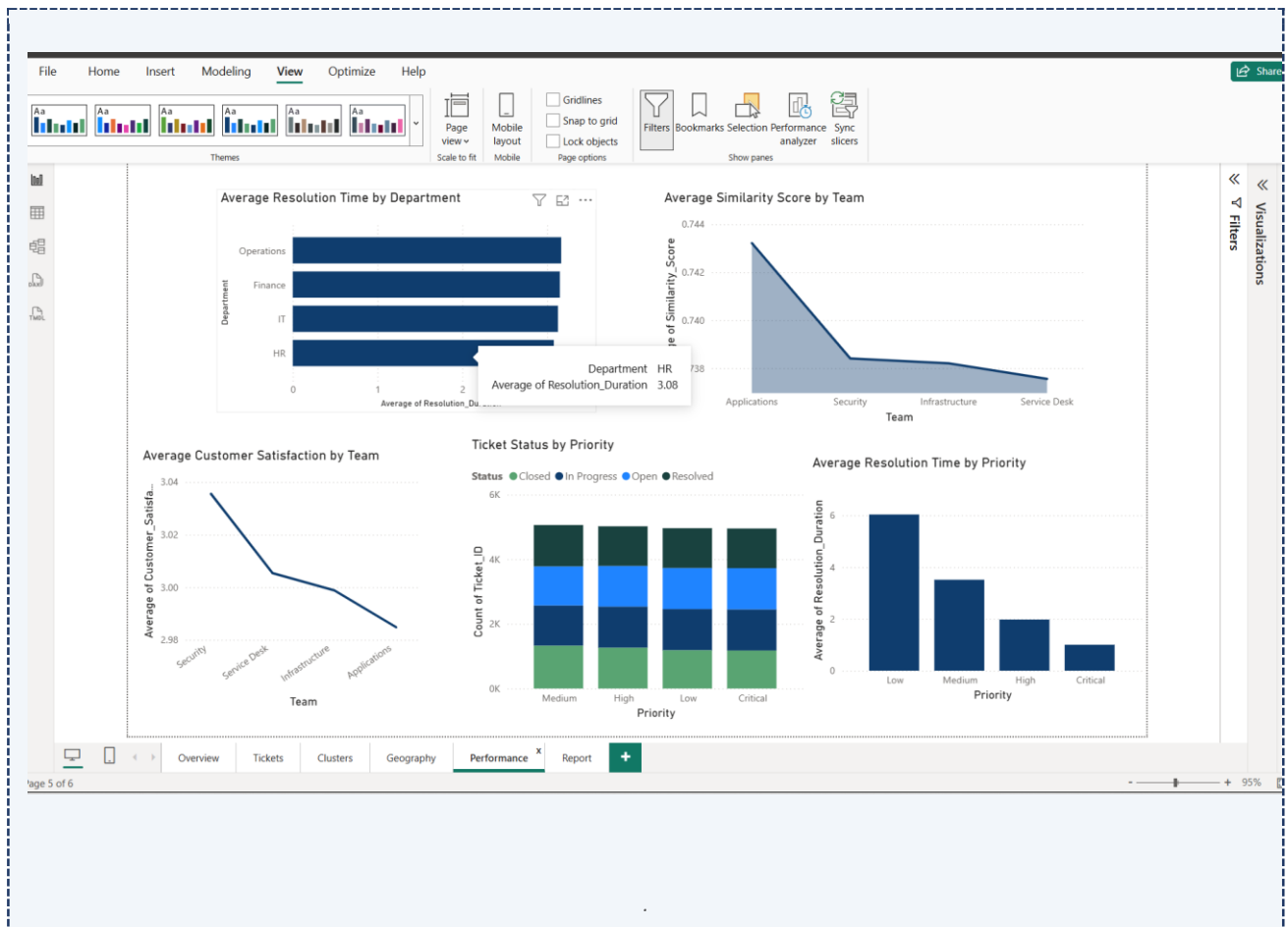


Figure 6.5



A native Power BI map visual was explored but was unavailable or disabled in the Power BI environment used for this project. The geographic analysis is therefore represented through the region- and country-based visuals above rather than a functional map.

Module 6 — Performance Analysis Outputs

Figure 6.6

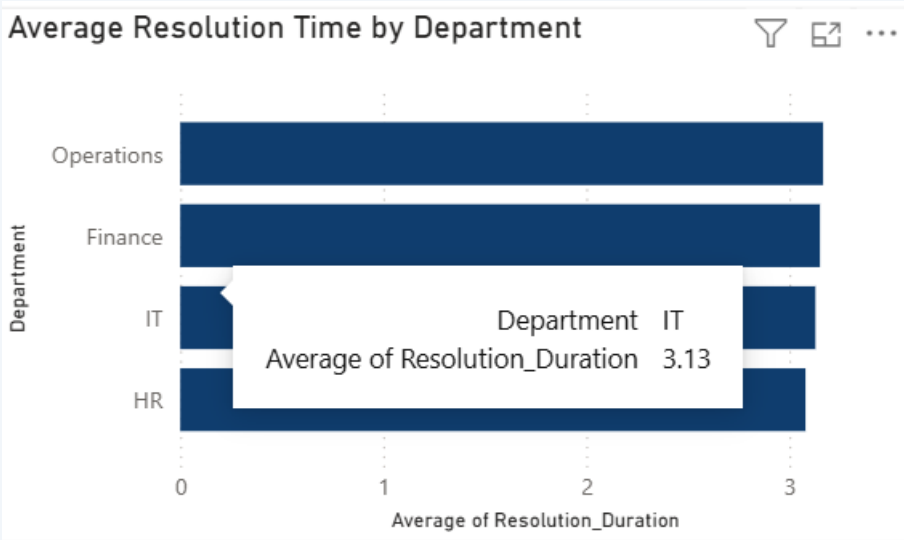


Figure 6.7

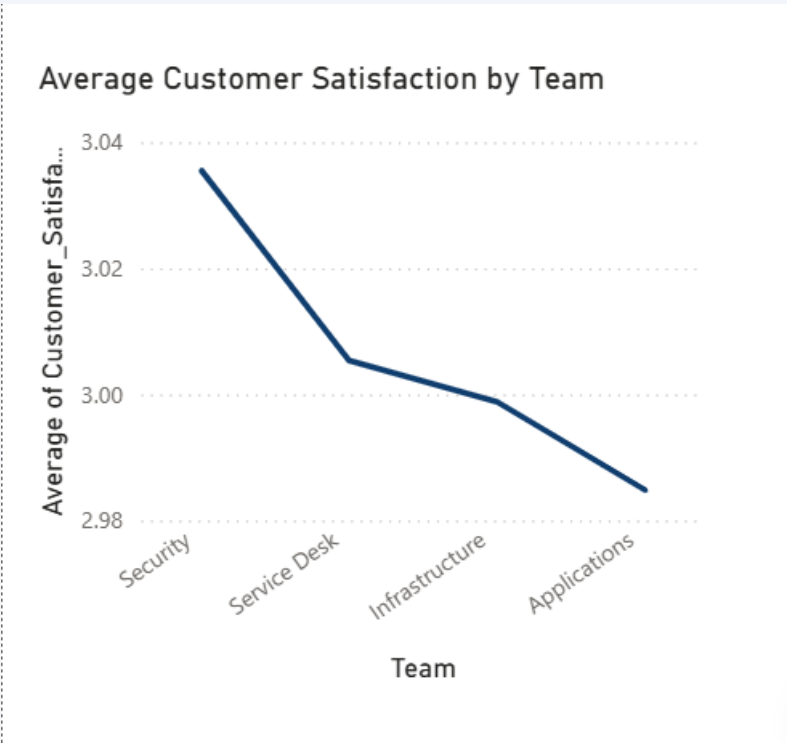


Figure 6.8

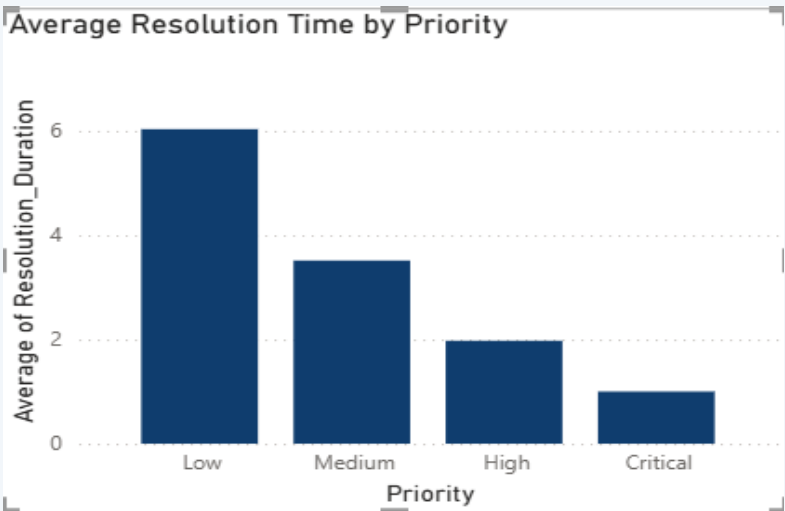


Figure 6.9

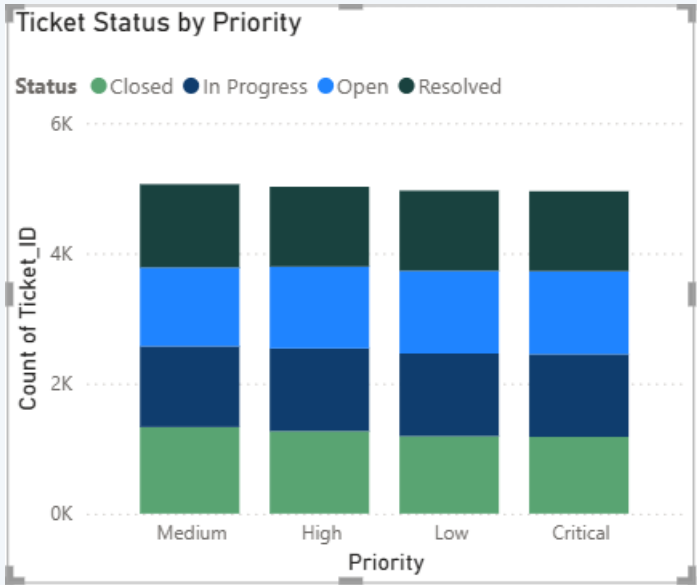
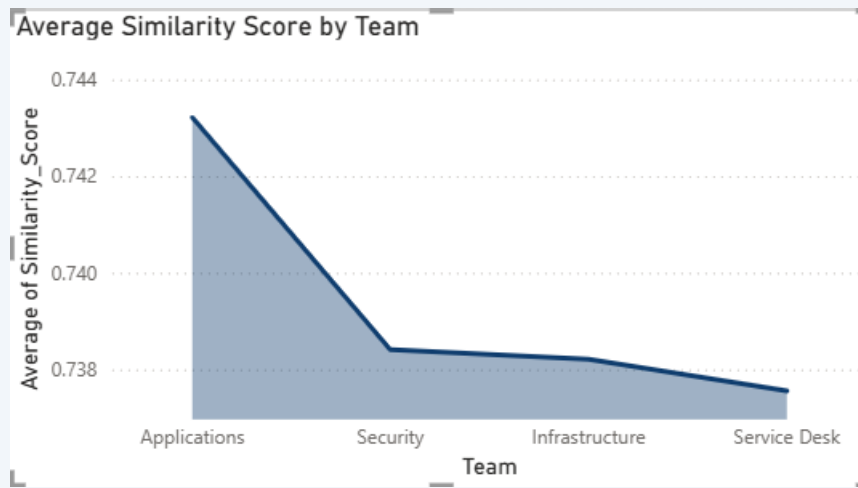


Figure 6.10



Module 7 — Power BI Dashboard & Report Preparation

Work completed during this milestone brought together the analytical work from Modules 2 through 6 into a single, organized and interactive Power BI report.

What Was Done

- The prepared Excel dataset was loaded into Power BI.
- Data fields were reviewed and confirmed field types were applied.
- The earlier Python-based analytical results were recreated as Power BI visuals.
- KPI cards were created for key headline measures.
- Visualizations were organized into six dedicated report pages.
- Dashboard navigation was structured across the pages.
- An executive report page was created to summarize overall findings.

Process Flow

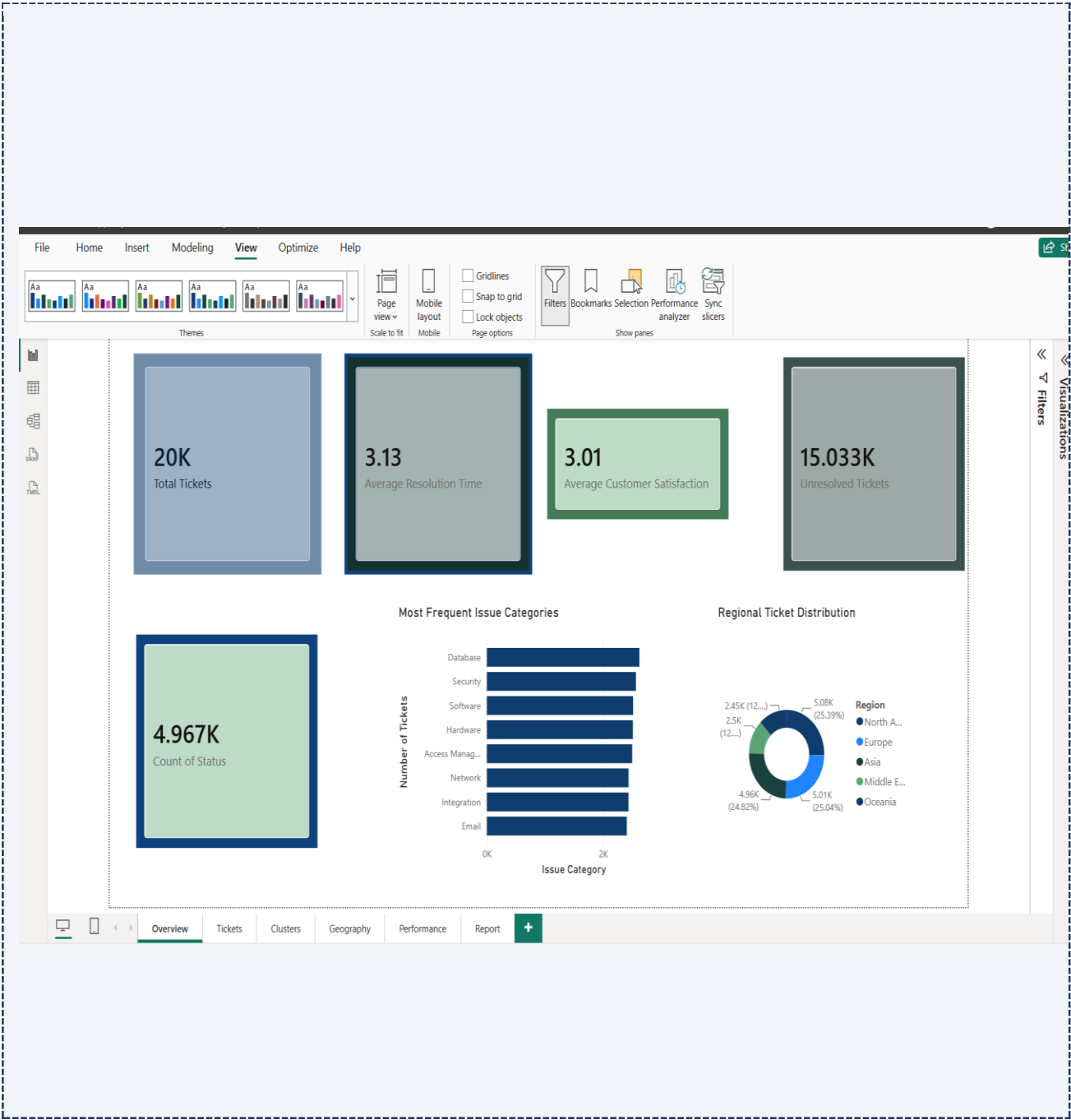
Excel Dataset → Analysis → Visualizations → Power BI → Interactive Dashboard → Executive Report

Report Structure

Page	Purpose
1. Overview	High-level KPI summary of overall support performance.
2. Tickets	Detailed ticket-level composition.
3. Clusters	Cluster and similarity analysis results.
4. Geography	Regional and country-level patterns.
5. Performance	Department, team and priority performance.
6. Report	Executive summary of key findings.

Power BI Page 1 — Overview

Figure 7.1 — Overview Page



Total Tickets: Total number of support tickets recorded — baseline measure of workload.

Average Resolution Time: Average time taken to resolve tickets — indicates overall efficiency.

Average Customer Satisfaction: Average satisfaction score across resolved tickets — indicates service quality.

Ticket Status Summary: Breakdown of tickets by status — shows open versus resolved workload.

Most Frequent Issue Categories: Categories generating the highest volume — highlights leading demand sources.

Regional Ticket Distribution: Ticket volume by region — shows geographic spread of activity.

Power BI Page 2 — Tickets

Figure 7.2 — Tickets Page



- Ticket Status Distribution:** Proportion of tickets in each status.
- Tickets by Support Channel:** Ticket volume by support channel.
- Ticket Priority Distribution:** Proportion of tickets by priority level.
- Priority Across Issue Categories:** Priority breakdown within each issue category.
- Ticket Type Distribution:** Proportion of tickets by ticket type.
- Issue Category Distribution:** Proportion of tickets by issue category.

Power BI Page 3 — Clusters & Similarity

Figure 7.3 — Clusters Page



Ticket Distribution Across Clusters: How tickets are distributed among clusters.

Average Similarity Score by Cluster: Average similarity score within each cluster.

Issue Category Distribution Across Clusters: Issue categories represented within each cluster.

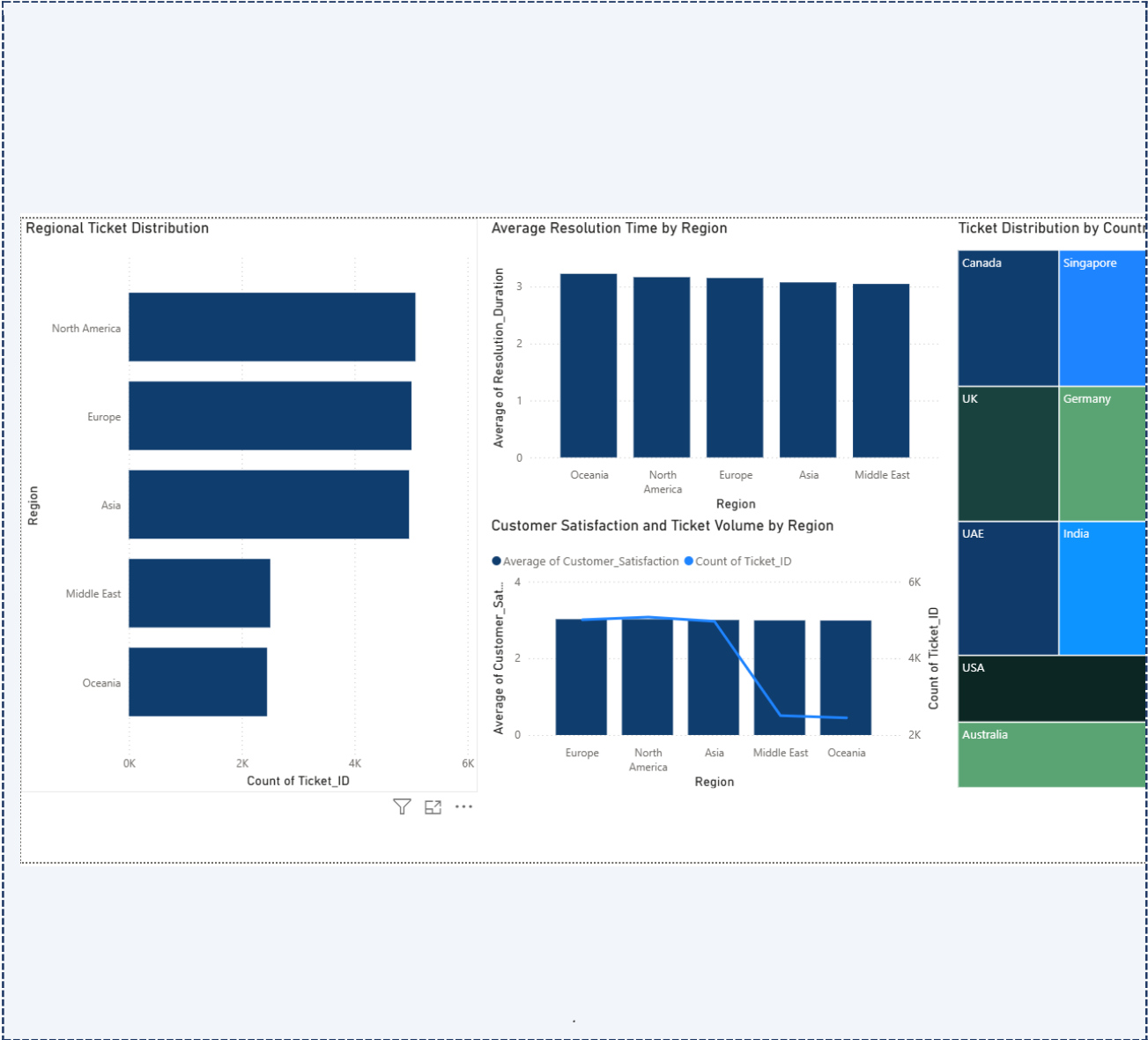
Ticket Volume by Cluster: Total tickets associated with each cluster.

Overall Cluster Similarity Index: Summary measure of similarity/coherence across all clusters.

Numerical similarity values are not stated here and should be confirmed from the completed dashboard.

Power BI Page 4 — Geography

Figure 7.4 — Geography Page



Regional Ticket Distribution: Ticket volume by region.

Average Resolution Time by Region: Average resolution duration by region.

Ticket Distribution by Country: Ticket volume at country level.

Customer Satisfaction by Region: Average satisfaction score by region.

Regional Performance: Combined comparison of regional support performance.

A functional Power BI map visual was unavailable/disabled in the environment used; the geographic analysis is represented through the region and country visuals above.

Power BI Page 5 — Performance

Figure 7.5 — Performance Page



Average Resolution Time by Department: Resolution duration compared across departments.

Average Customer Satisfaction by Team: Satisfaction compared across teams.

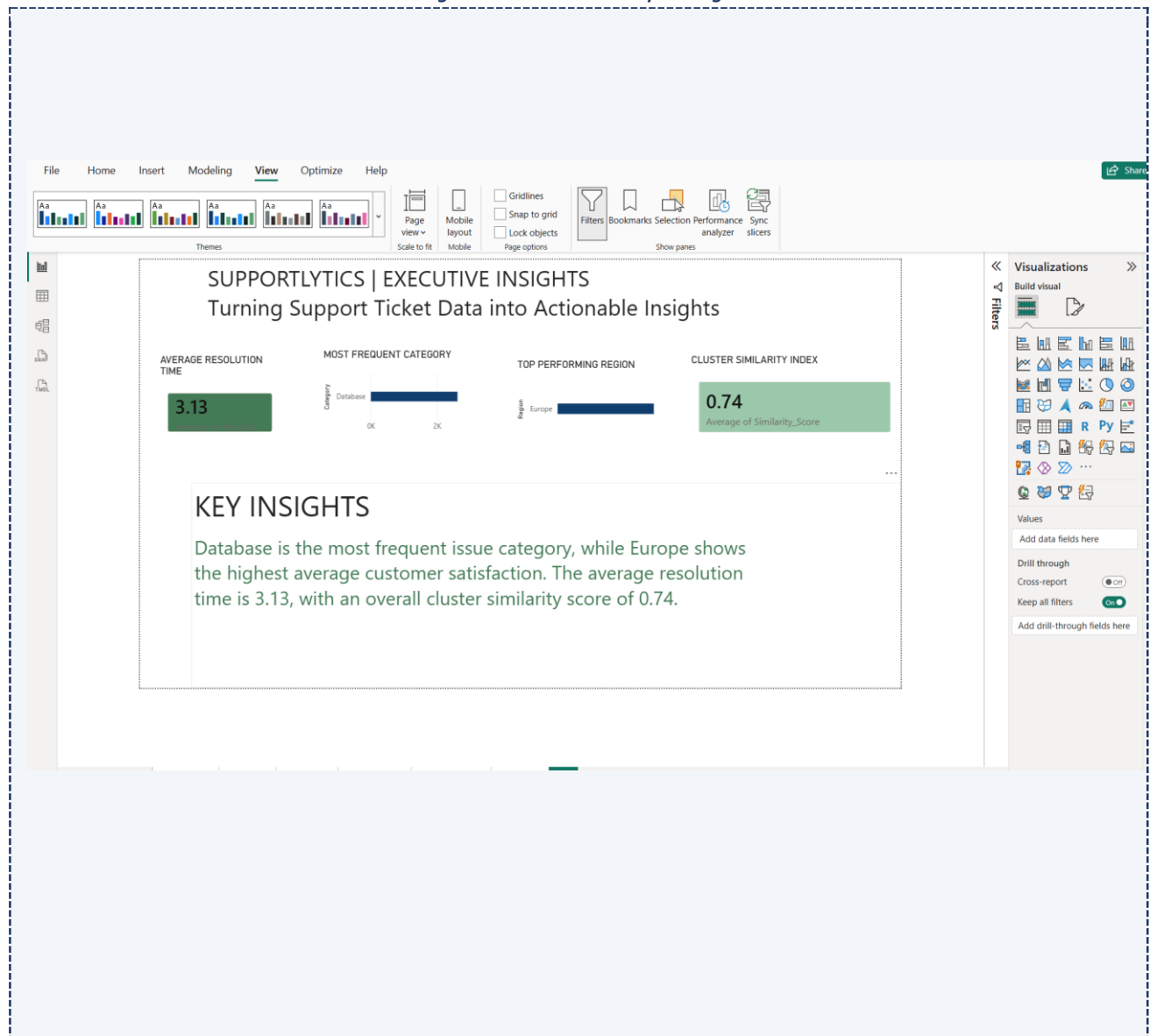
Average Resolution Time by Priority: Resolution duration compared across priority levels.

Ticket Status by Priority: Relationship between priority and current status.

Average Similarity Score by Team: Average similarity score for tickets handled by each team.

Power BI Page 6 — Executive Report

Figure 7.6 — Executive Report Page



Average Resolution Time: Overall average time taken to resolve tickets.

Most Frequent Category: Issue category with the highest ticket frequency.

Cluster Similarity Index: Overall similarity/coherence observed across clusters.

Top Performing Region: Region assessed as strongest by the selected performance measure.

Key Insights: Summary of the most important findings from the other five pages.

Numerical values are not invented here and should be inserted from the confirmed dashboard results.

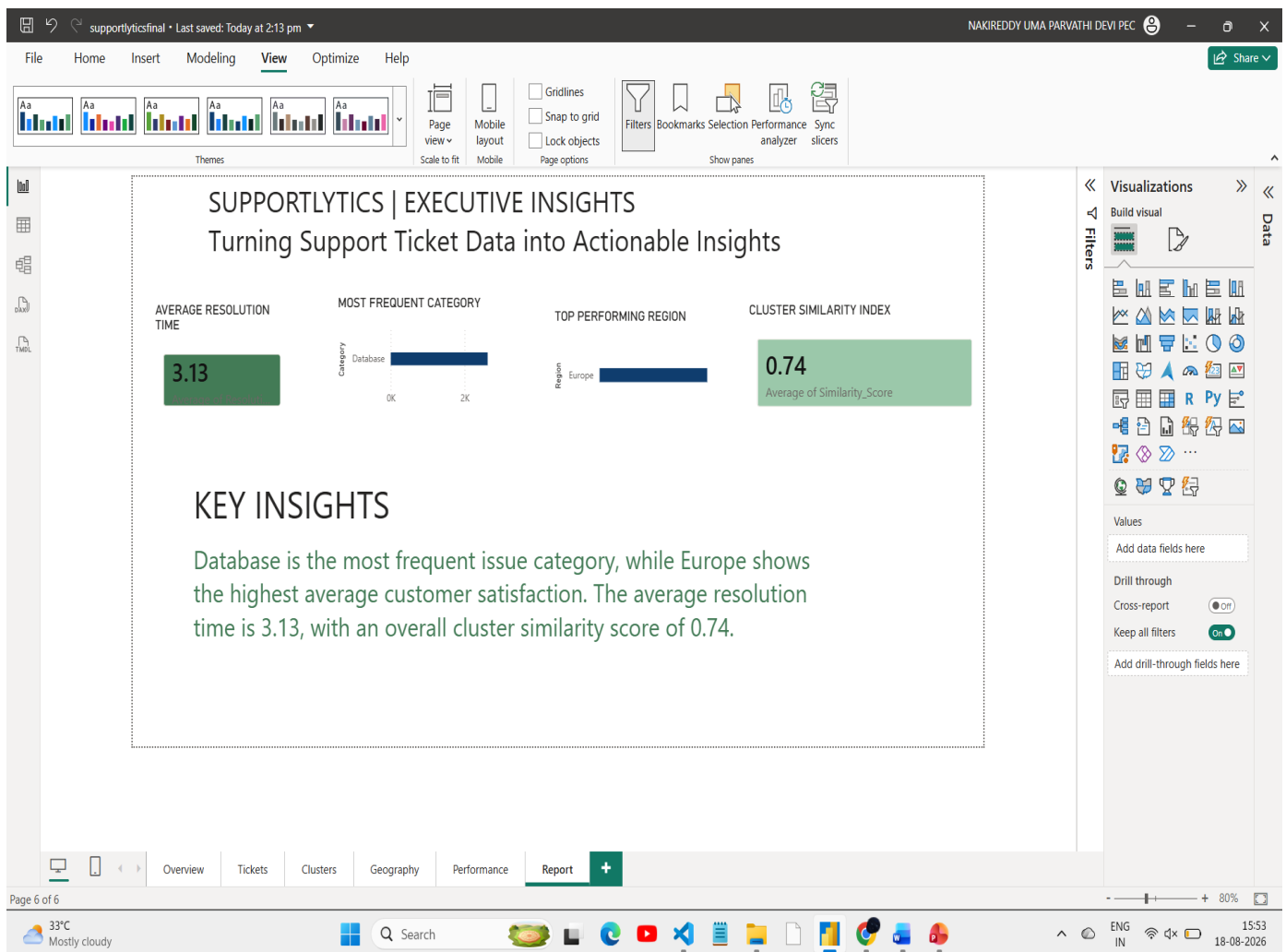
Module 7 — Dashboard Design & Final Contribution

The dashboard was divided into six pages to keep each analytical theme separately navigable while allowing the overall solution to be explored as a single interactive report.

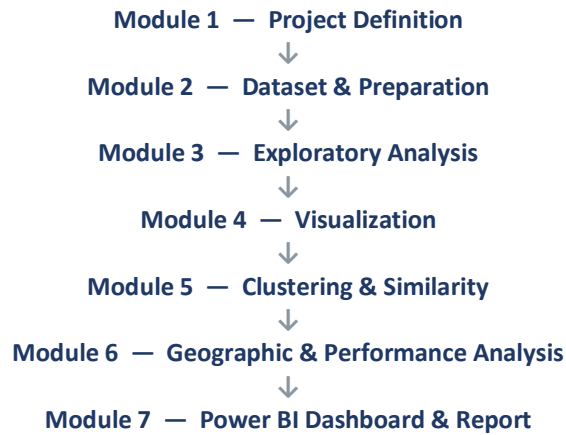
- KPI-based overview for a quick top-level summary.
- Detailed ticket analysis for status, channel, priority, type and category.
- A dedicated cluster and similarity analysis page.
- A dedicated geographic analysis page.
- A dedicated performance analysis page covering departments, teams and priorities.
- An executive summary page for stakeholders needing a concise view.
- Different chart types were used as appropriate to each analysis.
- A consistent layout and clear navigation were maintained across all pages.

Contribution to the Final Project

Module 7 transformed the analytical work completed in Modules 2 through 6 into one interactive, navigable Power BI report — the final deliverable of the Supportlytics project.



From Module 1 to Module 7



How Each Milestone Contributed

Module 1 defined the problem and scope. Module 2 prepared a reliable dataset. Module 3 established an understanding of the data's structure and main patterns. Module 4 turned those patterns into visual evidence. Module 5 identified recurring ticket groupings through clustering and similarity analysis. Module 6 extended the analysis geographically and by performance dimension. Module 7 consolidated all of this work into a single interactive Power BI dashboard and executive report, completing the transformation of raw ticket data into an actionable analytics solution.

Key Insights & Recommendations

Key Insights

The final dashboard is designed to allow users to identify the following, based on the confirmed results once available:

Ticket workload

Major issue categories

Priority patterns

Resolution performance

Customer satisfaction

Cluster patterns

Geographic differences

Team/department performance

Recommendations

- Focus resources on high-volume issue categories.
- Monitor unresolved high-priority tickets.
- Investigate areas with longer resolution times.
- Study recurring patterns identified through clustering.
- Use customer satisfaction results to guide improvement.
- Monitor regional and team-level performance on an ongoing basis.
- Use the Power BI dashboard for ongoing support-performance monitoring.

Project Outcome, Limitations & Conclusion

Project Outcome

Raw Data → Preparation → Analysis → Visualization → Clustering → Geographic Analysis → Performance Analysis → Power BI → Executive Report

Across the seven project milestones, the raw support-ticket dataset was progressively transformed into a structured, interactive analytics solution capable of supporting ongoing, data-driven decision-making.

Limitations

- The Power BI map visual was unavailable/disabled in the environment used, so geographic analysis was represented through region- and country-based visuals instead.

Conclusion

Supportlytics demonstrates a complete, milestone-based development process — from problem definition through dataset preparation, exploratory analysis, visualization, clustering, geographic and performance analysis, culminating in an interactive Power BI dashboard and executive report.

Supportlytics demonstrates how support-ticket data can be transformed into meaningful visual insights and used as a foundation for data-driven support improvement.

Final Supportlytics Dashboard & Supporting Evidence

Figure 25.1 — Final Supportlytics Power BI Dashboard

